claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a20a8930cad8062ae.jsonl`
- SHA-256: `4d467510ff6c620845eacfd9e2914967693c965a826b19ff31575ab5def27534`
- Source modified: `2026-06-06T17:34:12+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:32:36.224Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"The paper detects rally boundaries using only court geometry and player pose features (not audio or shuttle-detector trajectory): a CNN/BiLSTM classifies the serve-ready/serving action to mark rally START, and shuttlecock-flight gradient analysis marks rally END."

Source: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15 (primary)
Supporting quote: "A technique has been presented to detect the start of service and end of rally that utilizes court details and poses to accurately identify the service action from other badminton actions... Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest which signifies the end of the rally."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:32:39.448Z)

I'll verify this claim about the Springer paper. Let me start by fetching the primary source.

3. assistant (2026-06-06T17:32:47.422Z)

Let me fetch the primary source to verify the claim.

4. user (2026-06-06T17:32:50.956Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15"
- prompt: "Extract the abstract and any details about the method used to detect rally start (service) and rally end. Specifically: (1) Does it use court geometry/court details? (2) Does it use player pose features? (3) Does it use a CNN and/or BiLSTM to classify serve-ready/serving action for rally START? (4) Does it use shuttlecock-flight gradient analysis for rally END? (5) Does it use audio at all? (6) Does it use a shuttle-detector trajectory (like TrackNet/WASB)? Quote exact phrases about the methodology."

5. assistant (2026-06-06T17:32:56.028Z)

The Springer page requires authentication. Let me search for the paper details through other channels.

6. user (2026-06-06T17:33:07.553Z)

Web search results for query: "badminton detect start of service end of rally court details poses gradients shuttlecock flight Springer 978-3-031-69073-0"

Links: [{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"},{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link","url":"https://link.springer.com/article/10.1007/s42979-025-03935-0"},{"title":"(PDF) Using the trajectory of the shuttlecock as a measure of performance accuracy in the badminton short serve","url":"https://www.researchgate.net/publication/329030049_Using_the_trajectory_of_the_shuttlecock_as_a_measure_of_performance_accuracy_in_the_badminton_short_serve"},{"title":"A Machine Learning Framework for Shuttlecock Tracking and Player Service Fault Detection | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-39059-3_5"},{"title":"Effectiveness of Advanced Tracking Models for Shuttlecock and Court Line Detection in Small-Scale Badminton Matches | SpringerLink","url":"https://link.springer.com/chapter/10.1007/978-3-031-97578-3_71"},{"title":"Automated Service Height Fault Detection Using Computer Vision and Machine Learning for Badminton Matches","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10747833/"},{"title":"Automatic rally detection and scoring","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10751601"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"}]

Based on the search results, I found relevant information about the Springer publication (ISBN 978-3-031-69073-0) you queried. Here's what the research covers:

Service and End of Rally Detection in Badminton Videos

The technique detects the start of service and end of rally, utilizing court details and poses to accurately identify the service action from other badminton actions.

Detection Methods:

Service Detection:
BiLSTM (Bidirectional Long Short Term Memory) and CNN (Convolutional Neural Network) models are used for detecting the start of service. A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of serving and receiving actions was arranged in a 2D format that was used as inputs to the models.

Rally End Detection:
Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest which signifies the end of the rally. In computing the end of the rally, the algorithm considers the missing shuttlecock flight coordinates and when the shuttlecock is out of camera view.

Performance Results:
The experimental results demonstrated the highest classification accuracy of approximately 95% for the BiLSTM model, with frame intervals of 4, while the CNN achieved similar accuracy at frame intervals of 5.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-06T17:33:10.292Z)

Web search results for query: "badminton service action detection court details poses gradients shuttlecock flight end of rally CNN BiLSTM"

Links: [{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"},{"title":"A

Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link","url":"https://link.springer.com/article/10.1007/s42979-025-03935-0"},{"title":"Real-Time Badminton Action Recognition based on Media Pipe and Motion Bidirectional Encoder Representation Transformer","url":"https://www.researchgate.net/publication/395750036_Real-Time_Badminton_Action_Recognition_based_on_Media_Pipe_and_Motion_Bidirectional_Encoder_Representation_Transformer"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Request PDF","url":"https://www.researchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"},{"title":"[PDF] Recognition of Badminton Action Using Convolutional Neural Network | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Recognition-of-Badminton-Action-Using-Convolutional-Rahmad-Sufri/fab4f2e98a08f588f931a4e5c93885ba612c305d"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"},{"title":"VideoBadminton: A Video Dataset for Badminton Action Recognition","url":"https://arxiv.org/html/2403.12385v1"}]

Based on the search results, I found relevant information about badminton service action detection and rally analysis:

Service Action Detection

A technique has been developed that utilizes court details and poses to accurately identify the service action from other badminton actions, using the characteristics of service action to obtain features arranged in a 2D format for a convolutional neural network (CNN) to detect the service and non-service.

Deep Learning Models Comparison

For detecting the start of service, both Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN) models have been used while gradient computation determined the end of the rally. The experimental results demonstrated the highest classification accuracy of approximately 95% for the BiLSTM model with frame intervals of 4, while the CNN achieved similar accuracy at frame intervals of 5.

Features and Input Data

A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of serving and receiving actions was arranged in a 2D format that was used as inputs to the models.

Shuttlecock Flight and Rally Detection

Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest which signifies the end of the rally. For the detection of the end of a rally, it is required that at least 80% of consecutive frames exhibit no motion of the shuttlecock to signify the end of the rally.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:33:16.707Z)

The search results provide substantial detail. Let me probe the most contentious part of the claim: whether the method uses NO shuttle-detector trajectory and NO audio. The claim says "not audio or shuttle-detector trajectory." But the rally-END detection clearly uses shuttlecock flight coordinates/gradients, which requires tracking the shuttlecock. Let me look at the companion paper and verify the details.

9. user (2026-06-06T17:33:19.510Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/article/10.1007/s42979-025-03935-0
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0"
- prompt: "Extract the abstract and methodology. Does this badminton rally detection method use: (1) court geometry/court details, (2) player pose features, (3) CNN/BiLSTM to classify the serve action for rally START, (4) shuttlecock-flight gradient analysis (shuttlecock coordinates/trajectory) for rally END, (5) audio, (6) a shuttlecock detector/tracker to get shuttle coordinates? Quote exact phrases."

10. user (2026-06-06T17:33:28.587Z)

Web search results for query: ""Service and End of Rally Detection in Badminton" shuttlecock detection TrackNet coordinates tracking method"

Links: [{"title":"TrackNetV2: Efficient Shuttlecock Tracking Network | Request PDF","url":"https://www.researchgate.net/publication/350935356_TrackNetV2_Efficient_Shuttlecock_Tracking_Network"},{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"},{"title":"A Machine Learning Framework for Shuttlecock Tracking and Player Service Fault Detection | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-39059-3_5"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"},{"title":"TrackNetV3: Enhancing ShuttleCock Tracking with ...","url":"https://people.cs.nycu.edu.tw/~yushuen/data/TrackNetV3.pdf"},{"title":"A Machine Learning Framework for Shuttlecock Tracking and Player Service","url":"https://norma.ncirl.ie/6618/1/akshaymenon.pdf"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"},{"title":"Automatic rally detection and scoring","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10751601"}]

Based on the search results, here's what I found about service and end of rally detection in badminton using shuttlecock tracking:

Key Methods and Techniques

TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock. TrackNet focuses on tracking badminton shuttlecocks and generates a corresponding heatmap for each video frame, indicating the shuttlecock's position. By taking multiple frames into account simultaneously, this approach enables the model to learn the features of the shuttlecock and its trajectory.

Coordinate Tracking and Refinement

A sliding window of size 7 is defined to calculate the quadratic curve of coordinates if there are more than three predicted coordinates in this window. Any detected coordinate more than 50 pixels away from the curve is removed. Finally, if the distance between the predicted coordinate and the adjacent quadratic curves is less than 5 pixels, a final curve is obtained, which is estimated from the predicted coordinates across 15 consecutive frames.

End of Rally Detection

For the detection of the end of a rally, it is required that at least 80% of consecutive frames exhibit no motion of the shuttlecock to signify the end of the rally. A rally is a sequence of one or more strokes starting with a serve move. Usually a rally ends when a ball or a shuttlecock ceases to be in play.

Performance Results

Two videos of 4 and 12 minutes comprising 12 and 23 consecutive points are used to determine the accuracy of the proposed algorithm, showing 88.6% accuracy for a total of 35 rally points.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:33:37.449Z)

I now have enough detail to evaluate the claim precisely. Let me analyze the components carefully.

The claim has multiple parts:
1. "detects rally boundaries using only court geometry and player pose features (not audio or shuttle-detector trajectory)" ? this is the strong negative assertion
2. "CNN/BiLSTM classifies the serve-ready/serving action to mark rally START" ? supported
3. "shuttlecock-flight gradient analysis marks rally END" ? supported

The critical issue: the claim says the paper uses court geometry/pose **"not... shuttle-detector trajectory"**, AND it simultaneously says "shuttlecock-flight gradient analysis marks rally END." These two parts are internally contradictory. The rally-END detection clearly DOES rely on shuttlecock flight coordinates/trajectory ? the paper "considers the missing shuttlecock flight coordinates," requires "80% of consecutive frames exhibit no motion of the shuttlecock," and the companion/related work uses TrackNet to "extract the flight trajectory of the shuttlecock." So the parenthetical "(not... shuttle-detector trajectory)" is contradicted by the paper's own END-detection method, which is shuttlecock-trajectory-based.

Also, the START features explicitly include "gradient" among the 26 per-frame features (distance, angle, gradient, movements) ? and gradient relates to shuttle/motion trajectory, not pure pose. The START is court+pose+motion based, but the END is squarely shuttlecock-trajectory based.

Let me verify the gradient feature detail in the START stage to be precise.

12. user (2026-06-06T17:33:46.784Z)

Web search results for query: ""Service and End of Rally Detection in Badminton" 26 features 12 frames distance angle gradient movements service receiving"

Links: [{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos - Monash University","url":"https://research.monash.edu/en/publications/a-comparative-analysis-of-deep-learning-models-and-gradient-compu"},{"title":"Detection of service activity in a badminton game | Request PDF","url":"https://www.researchgate.net/publication/261087356_Detection_of_service_activity_in_a_badminton_game"},{"title":"Automated service scene detection for Badminton game analysis Using CHLAC and MRA | Request PDF","url":"https://www.researchgate.net/publication/290557755_Automated_service_scene_detection_for_Badminton_game_analysis_Using_CHLAC_and_MRA"},{"title":"Badminton Video Analysis based on Spatiotemporal and Stroke Features","url":"https://www.researchgate.net/publication/317159692_Badminton_Video_Analysis_based_on_Spatiotemporal_and_Stroke_Features"},{"title":"Real-Time Badminton Action Recognition based on Media Pipe and Motion Bidirectional Encoder Representation Transformer","url":"https://www.researchgate.net/publication/395750036_Real-Time_Badminton_Action_Recognition_based_on_Media_Pipe_and_Motion_Bidirectional_Encoder_Representation_Transformer"},
{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature

Link","url":"https://link.springer.com/article/10.1007/s42979-025-03935-0"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Request PDF","url":"https://www.researchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"},{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"}]

Based on the search results, I found relevant research on service and end of rally detection in badminton videos. Here's what the research shows:

Overview

Badminton videos are commonly used by coaches and players to analyze performance and gain insights into each point played in a match, and it is time consuming to manually annotate the start of service and end of each rally.

Feature Extraction and Input Format

A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of a serving and receiving actions was arranged in a 2D format that was used as inputs to the models.

Detection Approach

For detecting the start of service, two different models were used: Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN) while the gradient computation determined the end of the rally.

End of Rally Detection Method

Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest which signifies the end of the rally. In computing the end of the rally, the algorithm considers the missing shuttlecock flight coordinates and when the shuttlecock is out of camera view.

The research demonstrates a comprehensive approach combining deep learning models for service detection with gradient-based methods for identifying when a rally concludes.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. assistant (2026-06-06T17:33:57.677Z)

I have sufficient evidence to render a verdict. Let me analyze the claim against the established facts.

What the paper actually does (well-established across multiple search returns):
- Rally START: BiLSTM/CNN classifies the service action using 12 frames × 26 features (distance, angle, gradient, movements) derived from court details and poses. ? matches claim
- Rally END: gradient-based analysis of the shuttlecock's FLIGHT ? "considers the missing shuttlecock flight coordinates," requires ~80% of consecutive frames show "no motion of the shuttlecock." This is shuttlecock-trajectory/coordinate based.
- No mention of audio anywhere. ? matches "not audio"

The flaw in the claim: The claim states the paper detects rally boundaries "using only court geometry and player pose features (**not audio or shuttle-detector trajectory**)." This is contradicted by the paper's own END-detection method, which explicitly relies on **shuttlecock flight coordinates / trajectory** ("missing shuttlecock flight coordinates," "no motion of the shuttlecock"). The END stage IS shuttle-trajectory-based ? exactly what the parenthetical denies. The claim is internally contradictory: its own clause "shuttlecock-flight gradient analysis marks rally END" describes a shuttle-trajectory method, while the parenthetical says "not... shuttle-detector trajectory."

So the parts about START (court+pose+CNN/BiLSTM) and the absence of audio are correct, but the sweeping framing that the whole method avoids shuttle-trajectory is wrong. The shuttlecock flight/coordinate tracking is essential to the END detection. The claim overreaches by extending "court geometry + pose only" to the entire pipeline.

This is a partial refutation: the supporting quote itself says "Gradients are used to detect the flight of the shuttlecock," which is a shuttle-trajectory signal. The quote does NOT support "not... shuttle-detector trajectory" ? it directly contradicts it.

14. user (2026-06-06T17:34:07.012Z)

Structured output provided successfully

15. assistant (2026-06-06T17:34:11.999Z)

Verdict: refuted=true (high confidence).

The claim is partially correct but overreaches and is internally contradictory. The START-detection part (court geometry + player pose, CNN/BiLSTM classifying the service action) and the "no audio" part are accurate. But the sweeping framing "(not... shuttle-detector trajectory)" is contradicted by the paper's own rally-END method, which is explicitly shuttlecock-trajectory/coordinate based ("missing shuttlecock flight coordinates," "no motion of the shuttlecock," shuttlecock out of camera view). The supporting quote itself ? "Gradients are used to detect the flight of the shuttlecock" ? describes a shuttle-trajectory signal and therefore refutes, rather than supports, the "not shuttle-detector trajectory" assertion.